# RIFAT SARWAR

Full Stack Developer | AI & Robotics | CSE Graduate @ IUT

## PROFILE

I try to take my work seriously, but not myself. Love to code. Fueled by curiosity and occasional panic.

## CONTACT


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## EDUCATION

**BSc in Computer Science & Engineering**
Islamic University of Technology (IUT)
2022 – 2026
**CGPA:** 3.78 / 4.00

## ACHIEVEMENTS

**6th Globally**
International Rover Challenge (IRC)
1st in Bangladesh • 1st in Science Task

**16th Globally**
European Rover Challenge (ERC)

**Competed**
University Rover Challenge (URC)

**2nd Place**
CodeRush Hackathon

## SKILLS

**App Development**
Flutter

**Web Development**
Next.js, React

**Backend Development**
FastAPI, Spring Boot

**Databases**
MySQL, PostgreSQL, MongoDB

**CI/CD**
GitHub Actions

**Deployment**
AWS, DigitalOcean, Vercel, App Stores

**Machine / Deep Learning**
CV, NLP, LLM fine-tuning, RAG

**Robotics & Drones**
MAVLink, ROS, ArduPilot, WebRTC

## WORK EXPERIENCE

**Star Technologies** — Mar 2025 – Present
*Senior Software Engineer*

- Develop **full-stack mobile applications** along with **web admin panels**.
- **Design and implement the entire development workflow**, from visualization to final production.
- Custom-built **real-time chat, notification and data analysis systems**, along with machine learning pipelines such as **RAG** and a **voice assistant**.
- **Key projects:**
  - **GenieA** – in collaboration with *Sylhet Metropolitan Police, BD*. Consists of a citizen-side app, a police-side app and admin panels. **10K+ downloads.** Play Store App Store
  - **Health Star** – all-in-one healthcare app. MVP built by me from scratch; currently under active development. Play Store App Store

**Star Watch** (under Star Technologies) — Nov 2025 – Present
*Project Manager / Developer*

- Building **drone systems from scratch**.
- Working with **MAVLink, ROS, ArduPilot** and **WebRTC** video streaming.
- Automation pipeline using **LiDAR, optical flow and sonar**, with **AI-based mission planning**.

**Megamind Tech** (Remote – Canada) — Feb 2024 – Jul 2024
*LLM Developer / ML Engineer*

- **Offline inference and fine-tuning** of large language models (LLMs).
- **Optimized deployment pipelines** for resource-constrained environments and contributed to **custom fine-tuning** workflows for task-specific applications.

## TEAM PROJECTS

**Project Aero** — 2024 – 2025
*Team Lead*

- **Flight control** and **navigation**.
- CV pipelines for **real-time object detection and tracking**.
- **3D mapping** of environments from drone footage.

**Project Altair** (IUT Mars Rover Team) — 2023 – 2024
*Software Sub-team Senior Developer*

- Competed internationally at **IRC** (6th), **ERC** (16th) and **URC**.
- Implemented **four-wheel drive** and **6-joint arm** control systems.
- **Inverse kinematics** for precise arm manipulation.
- **Object recognition** models trained on a **custom dataset**, used to drive object grasping.
- **Autonomous navigation** with real-time **obstacle avoidance**.